

TECHNOLOGY STACK

Date: 11 July 2026

Team ID: XXXXXX

Project Name: EduGenie – Google Gemini Powered Learning Assistant Maximum Marks: 2 Marks

Define Your Technology Stack

EduGenie – Google Gemini Powered Learning Assistant is developed using a modern and scalable technology stack consisting of frontend technologies, backend frameworks, AI models, databases, deployment platforms, and third-party APIs. These technologies were selected to provide high performance, maintainability, scalability, security, and an interactive learning experience for students and educators.

Technology Stack Details

S.No	Architecture		Deployment	Justification / Purpose
	Component / Layer	Technology Chosen	Deployment	
	Frontend / Client-Side	HTML5, CSS3, JavaScript	Version Control & Collaboration	Git & GitHub
	Backend / Server-Side	Python, FastAPI, Uvicorn	Third-Party APIs / Libraries	Google Generative AI SDK, Transformers, Hugging Face, Pydantic,
2	AI / Large Language Models	Google Gemini 1.5 Pro, LaMini-Flan-T5		
3	Database / Data Storage	JSON Files, Local Storage		Provides a responsive and user-friendly interface for students and teachers to interact with EduGenie through forms, buttons, and dynamic content.
4				Handles application logic, API development, request processing, AI integration, and response generation with high performance.
5	Cloud /	Google AI Studio, Uvicorn (Local		Performs concept explanation, question answering, quiz generation, text summarization, and personalized learning recommendations.

Stores user preferences, generated quizzes, summaries, learning history, and application configuration in a lightweight format.

Provides access to the Gemini API and enables local deployment of the EduGenie application for development and testing.

Supports version control, collaboration among team members, project tracking, and source code management.

Enables AI model integration, natural language processing, request validation, template rendering, and efficient

Architecture			
S.No	Component / Layer	Technology Chosen	Justification / Purpose
		Jinja2	backend development.